names, but that doesn't mean we fully understand how they're formed.

Because the surface temperature of Venus is almost 470°C (900°F), the rocks there are much closer to their melting points than are those at the surface of the Earth. Rocks begin to soften and flow at much shallower depths on Venus than on Earth. This is very likely the reason that many geological features on Venus seem to be plastic and deformed.

The planet is covered by volcanic plains and highland plateaus. The geological constructs include volcanic cones, probable shield volcanos, and calderas. There are many places where we can see that lava has erupted in vast floods. Some plains features ranging to over 200 kilometers in size are playfully called "ticks" and "arachnoids" (which translates roughly as "spiderlike things")—because they are circular depressions surrounded by concentric rings, while long, spindly surface cracks extend radially out from the center. Odd, flat "pancake domes"—a geological feature unknown on Earth, but probably a kind of volcano—are perhaps formed by thick, viscous lava slowly flowing uniformly in all directions. There are many examples of more irregular lava flows. Curious ring structures called "coronae" range up to some 2,000 kilometers across. The distinctive lava flows on stifling hot Venus offer up a rich menu of geological mysteries.

The most unexpected and peculiar features are the sinuous channels—with meanders and oxbows, looking just like river valleys on Earth. The longest are longer than the greatest rivers on Earth. But it is much too hot for liquid water on Venus. And we can tell from the absence of small impact craters that the atmosphere has been this thick, driving as great a greenhouse effect, for as long as the present surface has been in existence. (If it had been much thinner, intermediate-sized asteroids would not have burned up on entry into the atmosphere, but would have survived to excavate craters as they impact this planet's surface.) Lava flowing downhill does make sinuous channels (sometimes under the ground, followed by collapse of the roof of the channel). But even at the temperatures of Venus, the lavas radiate heat, cool, slow, congeal, and stop. The